



IMPLEMENTATION TOPIC #2 · SESSION 3

Geonovum Sensor Data Testbed 2026

— Geo Insights × Building Changes

Session 3 · 2026

RECAP

A quick reintroduction

Geo Insights

A climate adaptation startup, building **UrbanAdapt**. Our mission: give Dutch municipalities geospatial intelligence to plan, implement, and monitor climate adaptation based on **measured effect**, not just models.

The bigger ambition

We are building a national sensor network for fieldlabs, with two goals:

- 1 Real-time insight for municipalities into how measures actually perform
- 2 Labeled training data for a geospatial foundation model

Currently in talks with Province of South Holland and VNG Realisatie.

STATUS

A realistic view of progress

Connecting three fieldlabs within a single testbed period was an ambitious plan. We are being honest: it takes more time than expected, but the underlying infrastructure is in place and steadily growing.

The Green Village (TU Delft)

• LIVE

3 data sources connected

Diergaarde Blijdorp

• Nearly ready

Final step before onboarding

EUR Campus, Rotterdam

• In progress

A bit further out

Living Lab Binckhaven (Den Haag)

• Scheduled

September 2026

AMS Institute (Amsterdam)

• Scheduled

2 fieldlabs, Sept/Oct 2026

We are onboarding slower than we hoped, but each new location now demonstrably takes less work than the last. That is the whole point of this architecture.

LIVE DATA

From stub to three real data sources

TGV Office Lab

5 sensors

SOURCE Confluent Cloud Kafka,
Avro-encoded

SENSOR Bosch BME680 (multi-parameter)

MEASURES Temperature, humidity, CO2,
air pressure

TOPIC `tud_gv_office-lab-climate`

Also carries a Climate Davis (Vantage Pro2)
outdoor weather station via the same topic.

Ohnics air quality

46 sensors

SOURCE SamenMeten citizen science
network, Delft

PROTOCOL Polling via the `5min.json`
endpoint every 5 minutes

FILTER Only sensors with the "de-" prefix
(Delft)

Public citizen-science sensors, registered
automatically.

Levellog groundwater

3 sensors

SOURCE CARS Online OData API

AUTH OAuth2 client credentials

POLLING Every 15 minutes

MEASURES Groundwater level across
multiple installation locations

 Each source has its own polling or streaming adapter, but all three converge on the same internal format before being pushed to FROST.

What we ran into along the way

PROBLEM Avro union types from Kafka sometimes arrive as nested dicts (`{"float": 3.14}`) instead of plain values.

SOLUTION A dedicated value extractor that normalises every Avro union representation to a clean float, regardless of the underlying type.

PROBLEM Polling sources (Ohnics, Levellog) resend the same values on every cycle when there is no new reading, creating duplicate observations on FROST.

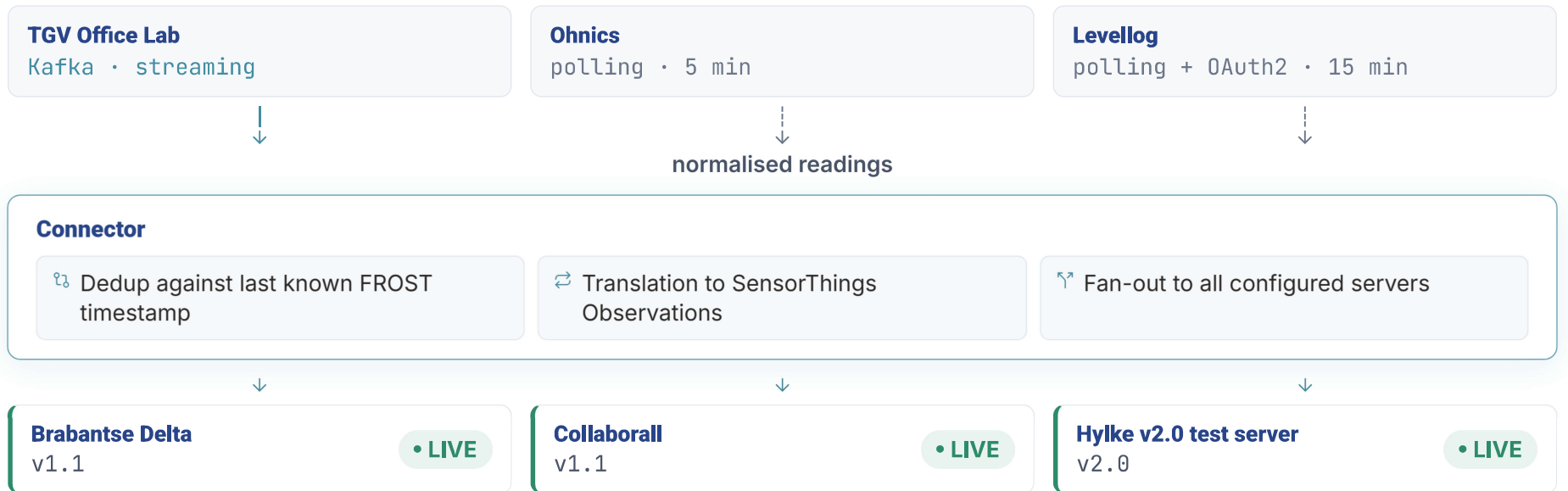
SOLUTION Dedup logic that reads the last known timestamp per sensor back from FROST itself before pushing. Only genuinely new readings are sent.

PROBLEM Feeding three servers at once (Brabantse Delta v1.1, Collaborall v1.1, Hylke's v2.0 test server), each with its own auth and sometimes a different API version.

SOLUTION A multi-target configuration where each server has its own URL, API version, and credentials. The connector fans out to every configured target on each push.

ARCHITECTURE

How it all comes together now



↗ **Monitoring module** direct push, bypasses FROST polling delay

PRODUCT

UrbanAdapt Monitoring Module

Since session 2 we expanded our existing monitoring module with features inspired by Justin's RIME Observatory, shown in the previous session.

CORE FEATURES

- 📄 Map view with live sensor values per location (3D, building heights)
- 🔄 Live updates without manual refresh
- 📍 Move and edit sensors directly on the map
- 📈 Time-series charts per sensor
- 📊 Project KPI dashboard

GEO INSIGHTS-SPECIFIC ADDITIONS

- 🗺️ Multi-project overview with health indicators (green: reporting, amber: partial, red: anomaly)
- 📁 FROST server selector per project
- ⊕ Sensor onboarding directly from the interface
- 📄 Description and photo per sensor, added at The Green Village's request
- 👤 Role-based access per stakeholder (municipality, fieldlab owner, researcher)

💡 We're now using The Green Village's feedback on these additions to guide the next round of features.

LIVE DEMO

Live demo: UrbanAdapt Monitoring Module

Map, live sensor values, project overview

TAKEAWAYS

From proof of concept to a production-ready pattern

- ✂ Three different protocols (Kafka/Avro, REST polling, OAuth2 API) prove the connector is generic enough for very different fieldlab infrastructure.
- ↗ Multi-server fan-out with per-server version and auth answers the session 2 question: sensors don't need to be divided across servers, they can go to every relevant server at once.
- 🛡 Dedup architecture makes repeated polling safe without polluting the data.
- ↗ The monitoring module is still a demo project, but it's advancing rapidly — we'll keep onboarding sensors at a steady, sustainable pace.

DISCUSSION

Where we're still stuck

1 **Scaling polling intervals**

Ohnics every 5 minutes, Levellog every 15. With dozens of sources: how do we prevent the connector from becoming overloaded, or FROST from receiving too many concurrent writes?

2 **Quality differences between sources**

A Kafka stream is near-real-time, an OAuth2 polling API is not. Should this difference be visible in the data itself, for example via resultQuality or a separate metadata flag?

3 **For Hylke specifically: what should we test on the v2.0 server?**

We are pushing to it now, but haven't yet run a structured comparison of v1.1 versus v2.0 behaviour. What would be most useful for you to see validated on your server?

ROADMAP

What's coming

COMING WEEKS	Blijdorp fully live
COMING MONTH	EUR Campus onboarding
SEPTEMBER	Living Lab Binckhaven
SEPTEMBER/OCTOBER	AMS Institute, two fieldlabs
ONGOING	Updating the how-to guide with lessons from three real data sources

We are taking the time to do this properly. Every new location needs to prove the pattern is reusable, not just that it works once.

SESSION 3 · CLOSING

Live on two servers.

101

datastreams live

2

data sources live

2

servers fed

Discussion and questions welcome



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